

ATOMIC STRUCTURE REVIEW

Parts of Atoms:

Most people already know that the atom is made up of three main parts, the protons and neutrons in the **nucleus** and the electrons somewhere outside of the **nucleus**.

Let's summarize:

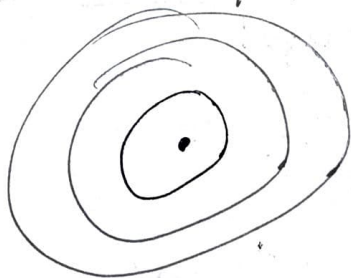
	proton	neutron	electron
symbol	p^+	n	e^-
charge	$+1$	0	-1
location	nucleus	nucleus	cloud (orbitals)
mass	1	1	0.511
size (see below)	10^{-15}	10^{-15}	10^{-18}

Let's make this more visual using information from the Chart of Fundamental Particles.

If the proton were 10 cm in diameter... the size of an orange, how big would everything be?

object	actual size	model size	model
proton	10^{-15} m	10 cm	orange
neutron	10^{-15} m	10 cm	orange
electron	10^{-18} m	0.01 cm	fly / grain
atom	10^{-10} m	10 km	

The atom is often represented as a miniature solar system. Draw it:



The **mass** of the atom is due to the nucleus

The **size** of the atom is due to the e^- cloud

How Many Particles in Each Atom?

The particle that defines the identity of an atom is the protons. (shown on the periodic table)

Every hydrogen atom has 1 proton.

Every magnesium atom has 12 protons.

Any atom that has 23 protons is ✓.

Any atom that has 92 protons is ✓.

The mass of an atom is mostly from the protons and neutrons.

Find O on the periodic table. It's mass is 16 amu. It has 8 protons. It must have 8 neutrons.

Electrically neutral atoms (as opposed to ions) have one electron for every proton.

Fill in this chart for these neutral atoms:

Atom	Mass	protons	neutrons	electrons
He	4	2	2	2
Si	28	14	14	14
Be	9	4	5	4
H	1	1	0	1
Rn	222	86	138	86
Ar	40	18	22	18
F	19	9	10	9
Pb	207	82	125	82

If the mass is not close to a whole number, it is because the atom has several isotopes. These are atoms with the same number of protons but different numbers of neutrons.

Chlorine has two isotopes: Cl-35 (17 p+ & 18 n⁰) and Cl-37 (17 p+ & 20 n⁰).

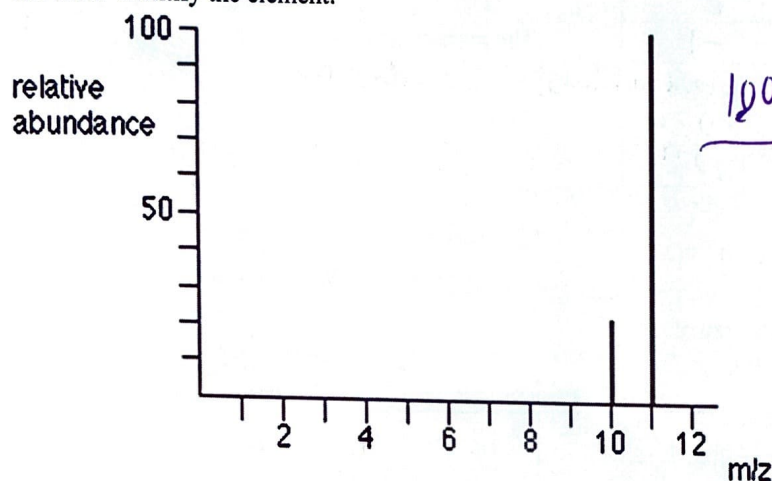
Practice with Average Atomic Mass Spectrum

1. Silver has two isotopes, Ag-107 (relative isotopic mass is 106.91) and Ag-109 (relative isotopic mass is 108.90). Ag-107 is 51.84% abundant in nature while Ag-109 is 48.16%. Calculate the average atomic mass of silver.

$$AM = 0.5184 \cdot 106.91 + 0.4816 \cdot 108.90$$

$$= 107.87 \text{ amu}$$

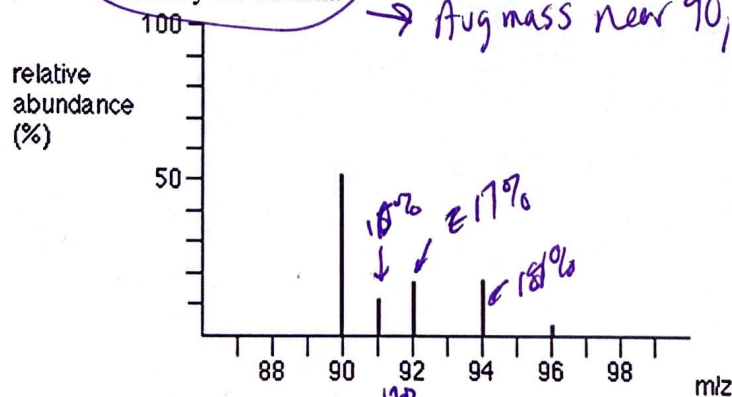
2. Below is the mass spec data for an element. Determine the percent abundance and the average atomic mass to this data. Identify the element.



$$\frac{100 \cdot 11 + 25 \cdot 10}{125} =$$

$$10.8 \text{ amu}$$

3. Below is the mass spec data for an element. Determine the percent abundance and the average atomic mass to this data. Identify the element.



→ Avg mass near 90, but larger!

Zr →

91 amu

Light, energy, wavelength, frequency, and atoms

Formulas and Constants

$$c = \lambda \nu$$

$$\nu = \frac{c}{\lambda}$$

$$\lambda = \frac{c}{\nu}$$

$$E = h\nu$$

$$E = \frac{hc}{\lambda}$$

$$c = 2.998 \times 10^8 \text{ m/s} \quad h = 6.626 \times 10^{-34} \text{ J}\cdot\text{s}$$

$$1 \text{ nanometer} = 1 \times 10^{-9} \text{ m} \quad 1 \text{ meter} = 1 \times 10^9 \text{ nm}$$

1. Perform the following conversions:

$$519 \text{ nm} = 5.19 \times 10^{-7} \text{ m}$$

$$2.12 \times 10^{-9} \text{ m} = 2.12 \text{ nm}$$

$$4 \times 10^{-6} \text{ m} = 4 \times 10^3 \text{ nm}$$

$$46700 \text{ nm} = 4.67 \times 10^{-5} \text{ m}$$

2. List all electromagnetic radiations from low energy to high.

radio	micro	IR	Visible	UV	Xray	gamma
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3. An FM radio station has a frequency of 88.9 MHz (1 MHz = 10^6 Hz, or cycles per second). What is the wavelength of this radiation in meters?

$$c = \lambda \nu$$

$$\lambda = 3.37 \text{ m}$$

4. The U.S. Navy has a system for communicating with submerged submarines. The system uses radio waves with a frequency of 76 s^{-1} . What is the wavelength of this radiation in meters? ~~In miles?~~

$$c = \lambda \nu \quad 3.00 \times 10^8 = \lambda \cdot 76$$

$$\lambda = 3.9 \times 10^6 \text{ m}$$

English units are ridiculous.

5. Violet light has a wavelength of about 410 nm. What is its frequency? Calculate the energy of one photon of violet light. What is the energy of 1.0 mol of violet photons?

$$c = \lambda \nu$$

$$3.00 \times 10^8 = 4.10 \times 10^{-7} \cdot \nu$$

$$\nu = 7.32 \times 10^{14} \text{ Hz}$$

$$E = h\nu = 6.626 \times 10^{-34} \cdot 7.32 \times 10^{14} = 4.8 \times 10^{-19} \text{ J}$$

6. The energy of a mole of photons of red light from a laser is 175 kJ/mol. Calculate the energy of one photon of red light. What is the wavelength of red light in meters? In nm? Compare the energy of photons of violet light with those of red light. Which is more energetic?

$$\frac{175 \text{ kJ}}{\text{mol}} \cdot \frac{1 \text{ mol}}{6.022 \times 10^{23}} = 2.91 \times 10^{-19} \text{ J}$$

$$E = \frac{hc}{\lambda}$$

$$2.91 \times 10^{-19} = \frac{6.626 \times 10^{-34} \cdot 3.00 \times 10^8}{\lambda}$$

$$\lambda = 6.84 \times 10^{-7} \text{ m}$$

$$= 684 \text{ nm}$$

$$\frac{4.8 \times 10^{-19} \text{ J}}{6.022 \times 10^{23}} = 7.97 \times 10^{-23} \text{ J/mol}$$

7. The most prominent line in the spectrum of neon is found at 865.438 nm. Other lines are found at 837.761 nm, 878.062 nm, 878.438 nm, and 1885.387 nm.

- (a) Which of these lines represents the most energetic light? Shortest! = 837.761 nm

- (b) What is the frequency of the most prominent line? What is the energy of one photon of this wavelength?

$$c = \lambda \nu$$

$$3.00 \times 10^8 = 8.377 \times 10^{-7} \cdot \nu$$

$$\nu = 3.58097 \times 10^{14} \text{ Hz}$$

$$E = h\nu$$

$$= 6.626 \times 10^{-34} \cdot 3.58097 \times 10^{14}$$

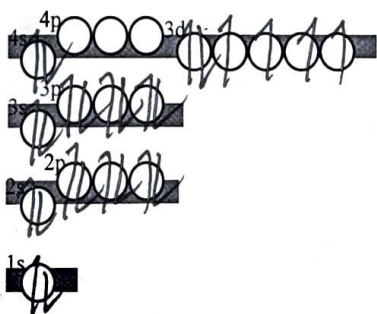
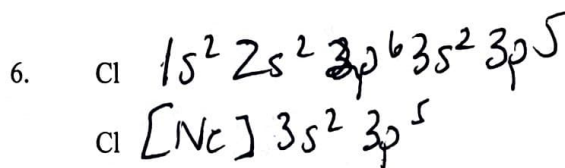
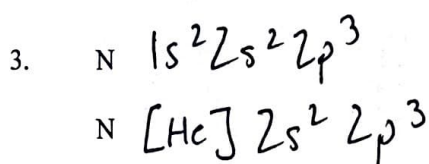
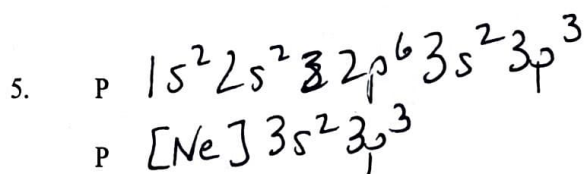
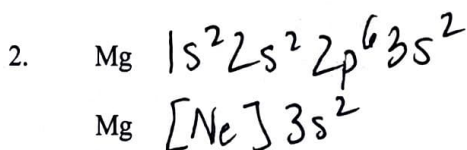
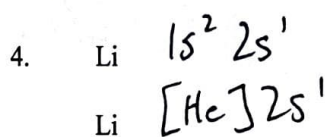
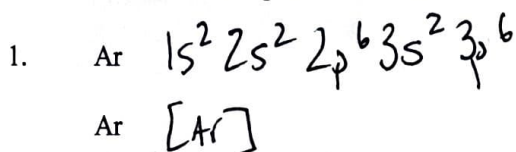
$$= 2.37275 \times 10^{-19} \text{ J}$$

WRITING ELECTRON CONFIGURATIONS

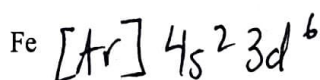
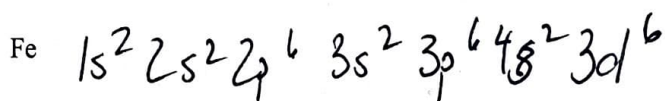
For each given element, fill in the orbital diagram and then write the electron configuration for the element.

1.	2.	3.	4.	5.	6.
Element: Ar	Element: Mg	Element: N	Element: Li	Element: P	Element: Cl
# of e ⁻ 's: <u>18</u>	# of e ⁻ 's: <u>12</u>	# of e ⁻ 's: <u>7</u>	# of e ⁻ 's: <u>3</u>	# of e ⁻ 's: <u>15</u>	# of e ⁻ 's: <u>17</u>

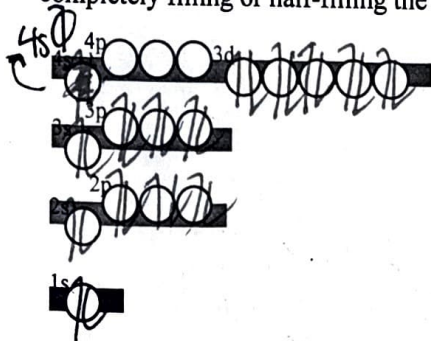
Write the electron configurations of each of these in **long form** and **short form**:



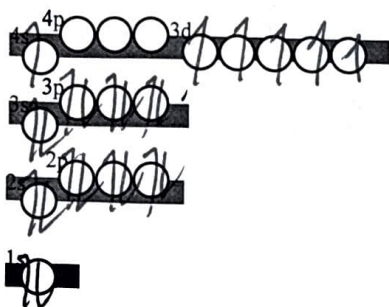
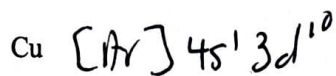
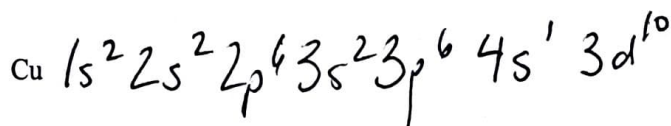
7. Fill in the orbital diagram for the element, Fe, and write the electron configuration of Fe in the long and short form.



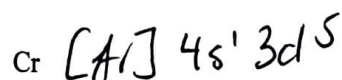
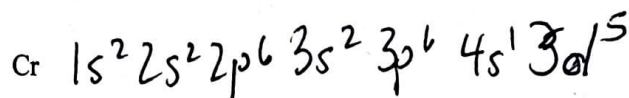
A few elements do not follow the "rules". There is some lowering of the energy of the atom by completely filling or half-filling the five d-orbitals.



8. Fill in the orbital diagram for the element, Cu, and write the electron configuration of Cu in the long and short form.



9. Fill in the orbital diagram for the element, Cr, and write the electron configuration of Cr in the long and short form.



Shade in the six elements that do not follow the Aufbau Principle:

Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn
Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd
La	Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg

1s	Fill in the orbitals that are filled by these elements.		1s
2s			2p
3s			3p
4s	3d		4s
5s	4d		5p
6s	5d		6p
7s	6d		7p

4f
5f

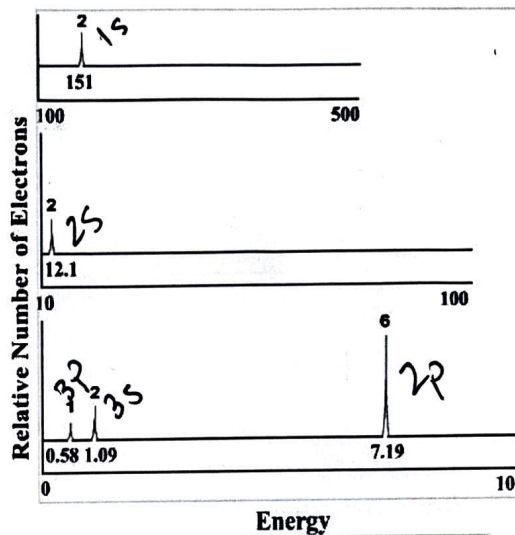
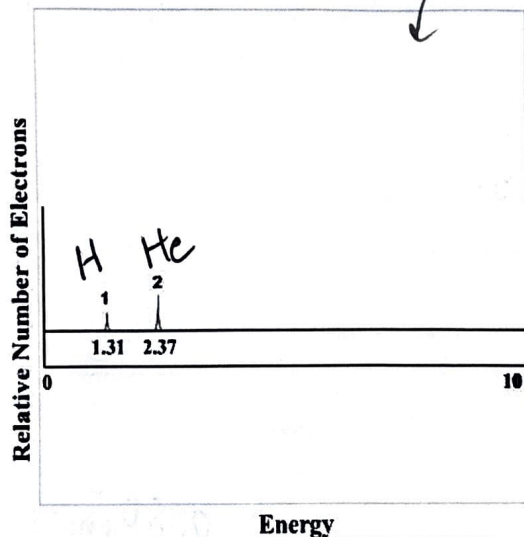
10. Write the orbital occupied by the last electron of each of the following elements:

As	W	Li	U	O	Rn	V
5p	5d	2s	5f	2p	6p	3d

Photoelectron Spectroscopy

Refer to the PES spectra below. For the spectrum on the left, there are two elements overlayed: the integration peak with 1 electron is that of hydrogen, whereas the other peak reflecting two electrons represents helium.

The spectrum on the right is used for questions 3-5.



For the left hand spectrum:

- How does this diagram show that the strength of attraction between protons and electrons as their amounts increase?

Stronger attraction in He (more E to remove 1s e⁻) b/c there are 2 protons in He vs 1p⁺ in H.

- What does this diagram suggest about the energy needed to remove an electron from a half-filled orbital versus a filled orbital?

Suggests it is less ... HOWEVER the reason for the difference is the #p⁺

For the right hand spectrum:

- Which energy value is associated with the 1s electrons? 151 units (Most energy to remove!)
- What element is represented by the right hand spectrum? 1s² 2s² 2p⁶ 3s² 3p¹ → Aluminum
- Explain the following questions:

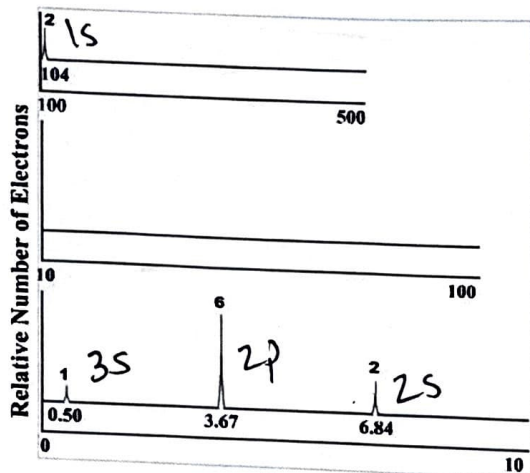
- Why does it take a significantly greater amount of energy to remove core electrons as compared to valence electrons?

They are closer to nucleus so attraction to p⁺ is greater than the valence e⁻

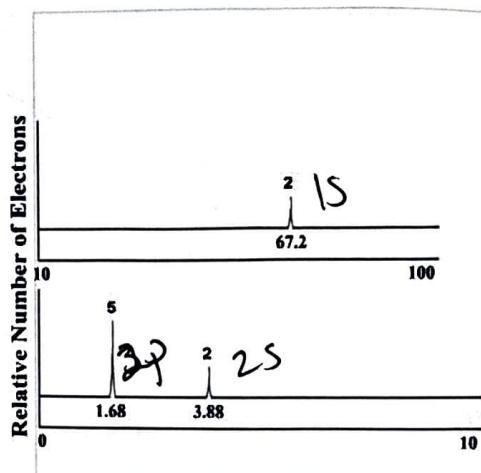
- Why do the integration peaks have different heights?

More e⁻ in an orbital means ~~more~~ higher probability of e⁻ being counted.

Refer to the PES spectra below. Identify the elements for each and then proceed to the prompts.



Na

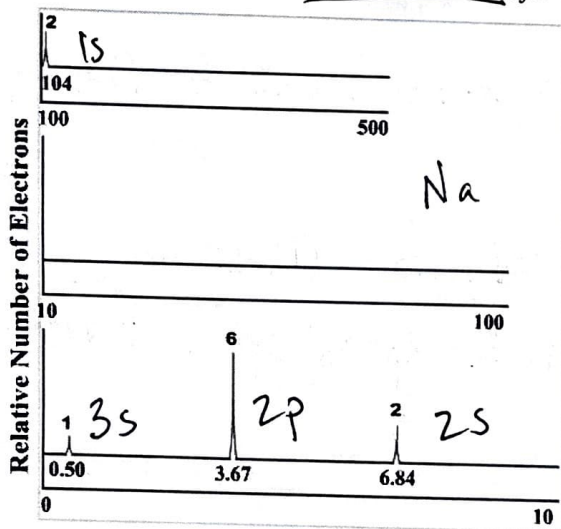


F

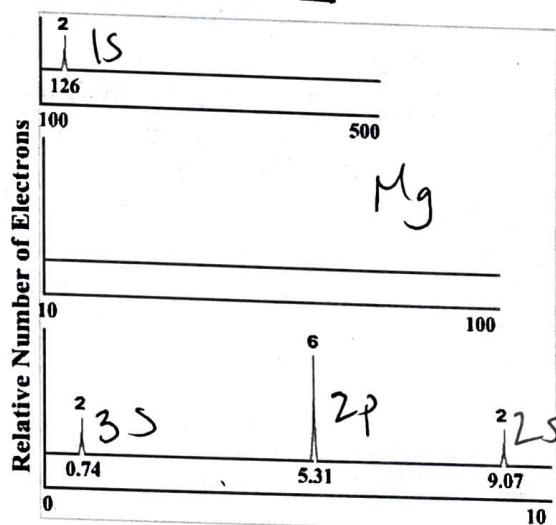
- 6) Based on the spectra above, discuss why the element in the left spectrum is better suited for forming a positive ion (cation), whereas the element in the right spectrum is better suited for forming a negative ion (anion).

Na has one valence e^- , removed w/ 0.50 units, forming a cation. F has 5 valence e^- , would require 1.68 to remove, so addition of e^- is preferred.

Refer to the PES spectra below. Identify the elements for each and then proceed to the prompts.



Na



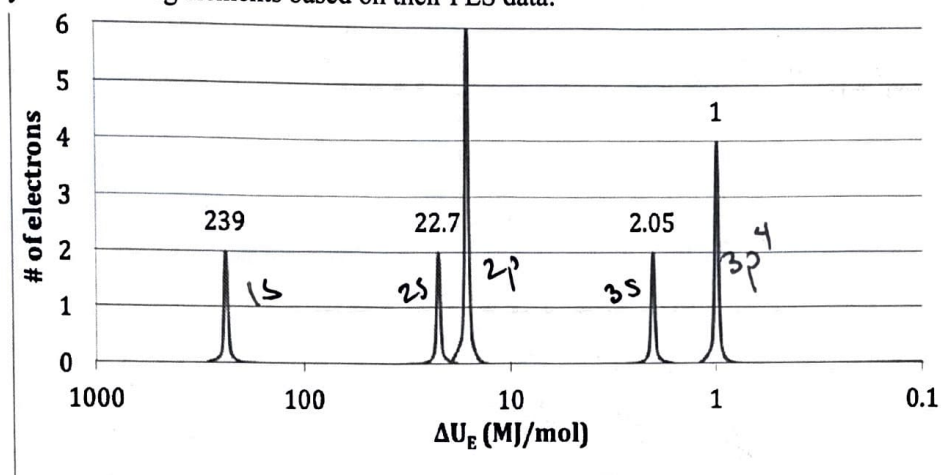
Mg

- 7) Based on the spectra above, which element is more likely to form a cation with a +2 charge? Why?

Mg can lose 2 valence e^- @ only 0.74 units each & will for Mg^{+2}

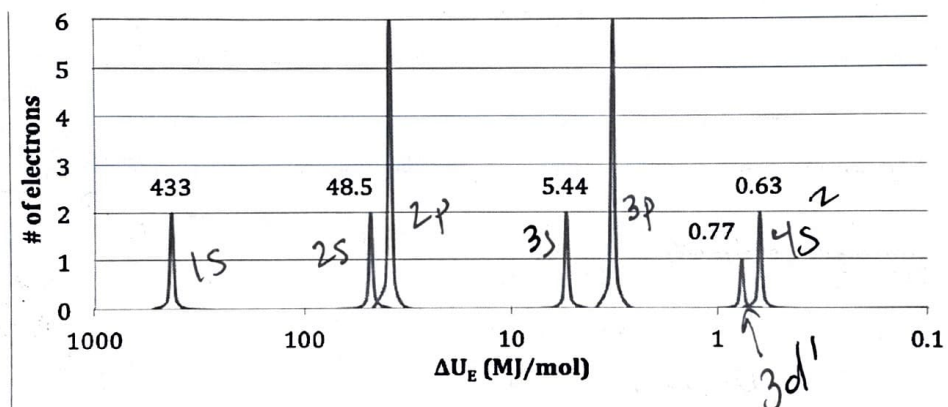
Na would lose 1 valence e^- @ only 0.50 each, but the 2nd e^- would require 3.67 units to remove. UNLIKELY

8) Identify the following elements based on their PES data:



a)

The element is: Sulfur



b)

The element is: Scandium

Detail needed for a 5,
not for a 3

Note:
d e⁻ lost after
~~Before~~ 4s
e⁻
When a Sc cation
is formed:
[Ar] 3d' is left
behind

Successive Ionization Energies

Removing one electron, and then another electron, and then another electron...

- 1st Ionization Energy: (IE_1) energy required to remove the first (highest energy) electron
- 2nd Ionization Energy: (IE_2) energy required to remove the second electron (second highest energy)
- Each additional electron requires **MORE** energy to remove than the previous one:

$$IE_1 < IE_2 < IE_3 \text{ (etc.)}$$

- Valence (outer) electrons require much **LESS** energy to remove than core (inner) electrons

$$IE_{\text{valence}} \ll IE_{\text{core}}$$

TABLE 8.1 Successive Values of Ionization Energies for the Elements Sodium through Argon (kJ/mol)

Element	IE ₁	IE ₂	IE ₃	IE ₄	IE ₅	IE ₆	IE ₇
Na	496	4560					
Mg	738	1450	7730				
Al	578	1820	2750	11,600			
Si	786	1580	3230	4360	16,100		
P	1012	1900	2910	4960	6270	22,200	
S	1000	2250	3360	4560	7010	8500	27,100
Cl	1251	2300	3820	5160	6540	9460	11,000
Ar	1521	2670	3930	5770	7240	8780	12,000

You can identify an element from the pattern of successive ionization energies!

FR Practice

IE ₁	IE ₂	IE ₃	IE ₄	IE ₅
801 kJ/mol	2,426 kJ/mol	3,660 kJ/mol	24,682 kJ/mol	32,508 kJ/mol

1. Which element from Period 2 does the table of successive ionizations energies above represent? Explain.

Boron B Since the increase between IE_3 & IE_4 is so large, this indicates the first 3 e^- removed were valence e^- , and the 4th was a core e^- . B has 3 valence e^- in the 2nd Period.

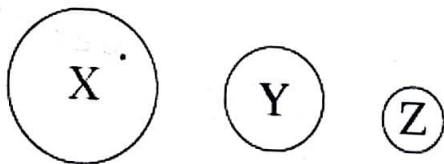
2. Would it require more energy to remove an electron from an F^+ ion or an F^{2+} ion? Justify your answer using Coulomb's Law.

F^{2+} b/c although the F^+ & F^{2+} ions have the same Z_{eff} , the F^{2+} has fewer e^- in the outer most shell & they experience less repulsion, allowing more attraction to nucleus

Multiple Choice Practice

	Ionization Energy
First	740 kJ/mol
Second	1,450 kJ/mol
Third	7,730 kJ/mol
Fourth	10,540 kJ/mol
Fifth	13,610 kJ/mol

- Given the table of ionization energies for the unknown element M shown above, which of the following is the most probable empirical formula for the compound element M would form with fluorine?
 - MF
 - ☒ MF₂
 - MF₃
 - MF₄
- Why does an ion of phosphorus, P³⁻, have a larger radius than a neutral atom of phosphorus?
 - There is a greater Coulombic attraction between the nucleus and the electrons in P³⁻.
 - The core electrons in P³⁻ exert a weaker shielding force than those of a neutral atom.
 - The nuclear charge is weaker in P³⁻ than it is in P.
 - ☒ The electrons in P³⁻ have a greater Coulombic repulsion than those in the neutral atom.
- Which of the following, sodium or magnesium, has greater metallic character and why?
 - Sodium, because sodium has fewer principal energy levels than magnesium.
 - ☒ Sodium, because sodium has a lower first ionization energy than magnesium.
 - Magnesium, because of the repulsion of magnesium's paired 4s electrons.
 - Magnesium, because magnesium has a greater effective nuclear charge than sodium.
- The diagram below shows the relative atomic sizes of three different elements from the same period. Which of the following statements must be true if trend exceptions are ignored?



- The effective nuclear charge will be the greatest in element X.
 - The first ionization energy will be greatest in element X.
 - The electron shielding effect will be greatest in element Z.
 - ☒ The electronegativity value will be greatest in element Z.
- Which of the following isoelectric species has the smallest radius?
 - S²⁻
 - Cl⁻
 - Ar
 - ☒ K⁺

6. Nitrogen's electronegativity value is between those of phosphorus and oxygen. Which of the following correctly describes the relationship between the three values?
- The value for nitrogen is less than that of phosphorus because nitrogen is larger, but greater than that of oxygen because nitrogen has a greater effective nuclear charge.
 - The value for nitrogen is less than that of phosphorus because nitrogen has fewer protons, but greater than that of oxygen because nitrogen has fewer valence electrons.
 - The value for nitrogen is greater than that of phosphorus because nitrogen has fewer electrons, but less than that of oxygen because nitrogen is smaller.
 - ☒ The value for nitrogen is greater than that of phosphorus because nitrogen is smaller, but less than that of oxygen because nitrogen has a smaller effective nuclear charge.
7. Which of the following, argon or krypton, has a higher IE_1 and why?
- ☒ Argon, because argon has fewer principal energy levels than krypton.
 - Argon, because argon has a larger effective nuclear charge than krypton.
 - Krypton, because krypton has more principal energy levels than argon.
 - Krypton, because krypton has a larger effective nuclear charge than argon.
8. The first five ionization energies for an element are listed in the table below.

First	Second	Third	Fourth	Fifth
8 eV	15 eV	80 eV	109 eV	141 eV

Based on the ionization energy table, the element is most likely to be

- Sodium
 - ☒ Magnesium
 - Aluminum
 - Silicon
9. Which of the following statements is true regarding sodium and chlorine?
- ~~Sodium has a greater electronegativity and a larger first ionization energy.~~
 - ~~Sodium has a larger first ionization energy and a larger atomic radius.~~
 - ~~Chlorine has a larger atomic radius and a greater electronegativity.~~
 - ☒ Chlorine has a greater electronegativity and a larger first ionization energy.
10. Consider the halogens chlorine and bromine. Which has a larger atomic radius and why?
- ~~Chlorine has a larger atomic radius because it has an increased number of principal energy levels.~~
 - ~~Chlorine has a larger atomic radius because it has a higher effective nuclear charge.~~
 - ☒ Bromine has a larger atomic radius because it has an increased number of principal energy levels.
 - Bromine has a larger atomic radius because it has a higher effective nuclear charge.

11. Which of the following species is NOT isoelectronic to Br^- ?

a. Se^{2-}

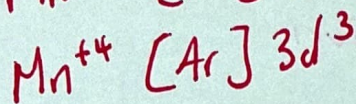
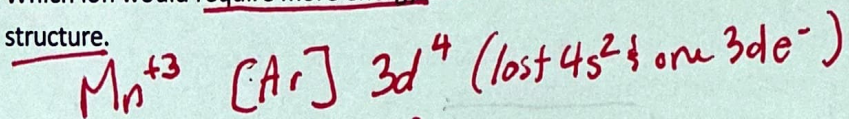
b. Kr

c. Rb^+

☒ d. K^+

Guided Practice: How will an answer be different when explained in terms of "atomic structure" vs "Coulomb's Law"?

1. Which ion would require more energy to remove an electron: Mn^{3+} or Mn^{4+} ? Explain using principles of atomic structure.



It would require more energy to remove an e^- from Mn^{4+} because although both have outer electrons removed from the same orbital ($3d$), the Mn^{4+} has fewer e^- in that orbital and there will be less e^-/e^- repulsion from the other $3d e^-$, making the attraction to the nucleus stronger.

2. Which ion would require more energy to remove an electron: Mn^{3+} or Mn^{4+} ? Be sure to reference Coulomb's Law in your explanation.

Mn^{4+} would require ~~less~~ more energy to remove because although the e^- are being removed from the same orbital ($3d$) and they experience the same effective nuclear charge (Z_{eff}), the e^- from Mn^{4+} have less e^-/e^- repulsion and will be closer to the nucleus, requiring more energy to remove.

Independent Practice

1. Using principles of atomic structure, explain why the Na^+ ion is larger than the Li^+ ion.
2. Which has a higher metallic character, an atom of lithium or an atom of boron? Explain your answer in terms of Coulomb's Law.

1.) Na^+ larger than Li^+ b/c Na^+ has a full $1s^2 \frac{1}{2} 2s^2 \frac{1}{2} 2p^6$ shell (2 full shells) while Li^+ only has a full $1s^2$ shell

2.) Metallic Character is measured by how easy an electron is lost by an atom, or how low the IE is. Li has a ~~lower~~ more metallic character than Boron. Even though the e^- for both are similar distances in the 2nd shell, Li has less Z_{eff} making the Coulombic attraction less than

3. Which would be greater for the aluminum atom, its first ionization energy or its second ionization energy? Explain using principles of atomic structure.

2nd e^-/e^- repulsion

4. Which element Cl or Br, has a greater first ionization energy? Justify your answer using Coulomb's Law.

Cl, e^- in 3p closer than 4p even though same z_{eff} .